

Simple Pendulum; Not so simple!

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The starting point for studying the wide variety of problems encountered in classical mechanics is none other than the simple pendulum. Although the problem may appear deceptively simple, I should warn you: looks can be very deceiving. Let us therefore take this seemingly simple system and study it carefully using the powerful framework of Lagrangian mechanics.

Consider a simple pendulum consists a bob of mass m is suspended from point P with the help of an inextensible string of length L . What we want to study is the dynamics of this system, i.e. how the position of the bob evolves as time progresses. In order to do this through the Lagrangian mechanics, we first have to find out the generalized coordinates and the by writing the Euler Lagrange equation we get the EOM.

We consider the motion of the bob restricted to the vertical plane, so that we can describe the position of bob as two coordinates (x, y) . But as the length of the string is fixed, its advantageous to do this problem in the plane polar coordinates where we can describe the position by an angle θ , which is measured from the downward vertical. This is schematically shown in Fig. 1.

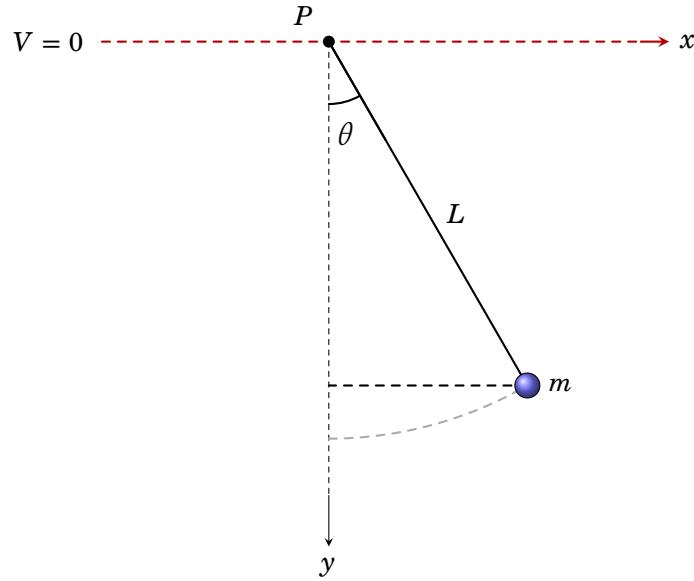


Figure 1: Simple Pendulum.

As everybody knows the coordinates are connected as :

$$x = L \sin \theta \quad (1a)$$

$$y = L \cos \theta \quad (1b)$$

Take the potential energy $V = 0$ at the suspension point P , which gives us the Lagrangian $\mathcal{L}$ as :

$$\mathcal{L} = \frac{m}{2} [\dot{x}^2 + \dot{y}^2] + mgy \quad (2)$$

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Now using Eq. (1) we get :

$$\dot{x} = L \cos \theta \dot{\theta} \quad (3a)$$

$$\dot{y} = -L \sin \theta \dot{\theta} \quad (3b)$$

Substituting Eqs. (1) and (3) in Eq. (2) we get :

$$\begin{aligned} \mathcal{L} &= \frac{m}{2} \left[\left(L \cos \theta \dot{\theta} \right)^2 + \left(-L \sin \theta \dot{\theta} \right)^2 \right] + mgL \cos \theta \\ &= \frac{mL^2 \dot{\theta}^2}{2} \left[\sin^2 \theta + \cos^2 \theta \right] + mgL \cos \theta \\ &= \frac{mL^2 \dot{\theta}^2}{2} + mgL \cos \theta \end{aligned} \quad (4)$$

Euler-Lagrange equation in angular variable θ becomes :

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) - \frac{\partial \mathcal{L}}{\partial \theta} = 0 \quad (5)$$

Using the Lagrangian obtained in Eq. (4) and inserting in Eq. (5) we get :

$$mL^2 \ddot{\theta} - (-mgL \sin \theta) = 0, \text{ or}$$

$$\boxed{\ddot{\theta} + \frac{g}{L} \sin \theta = 0} \quad (6)$$

Eq. (6) governs the time evolution of angular variable θ . Now if we use small angle approximation

$$\sin \theta \approx \theta$$

then Eq. (6) takes a very well known form :

$$\boxed{\ddot{\theta} + \omega^2 \theta = 0} \quad (7)$$

Where the quantity $\omega = \sqrt{\frac{g}{L}}$ is known as the angular frequency. Eq. (7) is known as SHM equation the solution of which is given by :

$$\theta(t) = A \sin \omega t + B \cos \omega t \quad (8)$$

A and B are the constants that can be found out given the proper initial conditions. And that's all.

Now comes to the another part of this article. Suppose in some universe or something you don't know about this plane polar coordinate system. All you know is the Cartesian coordinate system. Then how you are going to solve for the bobs trajectory. You are forced to use Euler Lagrange equation in coordinates (x, y) right! Then naively following Eq. (5) you might think to write the EOM as :

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) - \frac{\partial \mathcal{L}}{\partial x} = 0$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{y}} \right) - \frac{\partial \mathcal{L}}{\partial y} = 0$$

But this is horribly wrong. The question that comes naturally in ones mind what's the issue here ??? Let's first discuss about the degrees of the freedom of this system. Even though we need two coordinates to describe the position of bob (x, y) but as the length of the string is fixed, we have a constraint relation, i.e.

$$x^2 + y^2 = L^2 \quad (10)$$

Which reduces the coordinates from two one. So we can work in either x or y as we wish to. But the underlying physics will be the same that we will see very soon.

If you can remember, when we have constraints we have to use Lagrange's equation of mixed type ² which is given by :

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_k} \right) - \frac{\partial \mathcal{L}}{\partial q_k} = \sum_i^N \lambda_i \frac{\partial \phi_i}{\partial q_k} \quad (11)$$

Where q_k 's are the coordinates and we have N such holonomic constraint relations which are given as :

$$\phi_i(q_1, q_2, \dots, q_n, t) = 0, \quad i = 1, 2, \dots, N \quad (12)$$

Also λ_i 's are known as Lagrange's undermined multipliers and the term in the R.H.S. is known as the forces of constraint. Let's try to use this to solve for EOM. Now following Eq. (10) we have constraint :

$$\phi = x^2 + y^2 - L^2 = 0 \quad (13)$$

There is only one constraint and hence only one undetermined multiplier, call it λ . Using Eq. (11) we get :

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) - \frac{\partial \mathcal{L}}{\partial x} = \lambda \frac{\partial \phi}{\partial x}$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{y}} \right) - \frac{\partial \mathcal{L}}{\partial y} = \lambda \frac{\partial \phi}{\partial y}$$

²See https://en.wikipedia.org/wiki/Lagrangian_mechanics#Equations_of_motion

With a bit of algebra one can easily get :

$$m\ddot{x} = 2\lambda x \quad (15a)$$

$$m\ddot{y} - mg = 2\lambda y \quad (15b)$$

Looking at Eq. (15) we see that we want to solve for three unknowns (x, y, λ) . But we only have two such equations. Now how to overcome this issue. The answer lies in the fact that we have to use constraint relation Eq. (10) to substitute one variable either x or y and we can then proceed.

Differentiating Eq. (10) we get :

$$2x\dot{x} + 2y\dot{y} = 0 \quad (16)$$

Differentiating again we get :

$$x\ddot{x} + \dot{x}^2 + y\ddot{y} + \dot{y}^2 = 0 \quad (17)$$

Which gives us :

$$\ddot{y} = -\frac{x\ddot{x} + \dot{x}^2 + \dot{y}^2}{y}$$

Using Eq. (16) we substitute $\dot{y} = -\frac{x\dot{x}}{y}$ in last equation to obtain :

$$\begin{aligned} \ddot{y} &= -\frac{x\ddot{x} + \dot{x}^2 + \left(-\frac{x\dot{x}}{y}\right)^2}{y} \\ &= -\frac{x\ddot{x} + \dot{x}^2 \left[1 + \frac{x^2}{y^2}\right]}{y} \\ &= -\frac{x\ddot{x} + \frac{\dot{x}^2 L^2}{L^2 - x^2}}{\sqrt{L^2 - x^2}} \end{aligned}$$

So far so good. Now from Eq. (15b) we get :

$$\begin{aligned} 2\lambda &= \frac{m\ddot{y} - mg}{y} \\ &= \frac{m\ddot{y}}{y} - \frac{mg}{y} \\ &= -\frac{mx\ddot{x} + \frac{m\dot{x}^2 L^2}{L^2 - x^2}}{L^2 - x^2} - \frac{mg}{\sqrt{L^2 - x^2}} \end{aligned} \quad (18)$$

Substituting which in R.H.S. of Eq. (15a) gives :

$$\begin{aligned} m\ddot{x} = 2\lambda x &= -\frac{mx^2\ddot{x} + \frac{m\dot{x}^2 L^2}{L^2 - x^2}}{L^2 - x^2} - \frac{mgx}{\sqrt{L^2 - x^2}} \\ \Rightarrow (L^2 - x^2)m\ddot{x} &= -mx^2\ddot{x} - \frac{m\dot{x}^2 L^2}{L^2 - x^2} - mgx\sqrt{L^2 - x^2} \end{aligned}$$

$$\Rightarrow mL^2\ddot{x} + \frac{m\dot{x}^2L^2}{L^2 - x^2} + mgx\sqrt{L^2 - x^2} = 0$$

Finally we obtain the EOM in x given by :

$$\boxed{\ddot{x} + \frac{x\dot{x}^2}{L^2 - x^2} + \frac{gx}{L^2}\sqrt{L^2 - x^2} = 0} \quad (19)$$

Eq. (19) governs the evolution of the coordinate x and using the constraint relation Eq. (10) we can find out the other coordinate y at any instant t provided we know $x(t)$.

In terms of variable x this looks a bit tedious. Now one might think that the equation we obtained here and in Eq. (6) before are they same or not. To see this, we first evaluate :

$$\begin{aligned} x &= L \sin \theta \\ \dot{x} &= L \cos \theta \dot{\theta} \\ \ddot{x} &= L \cos \theta \ddot{\theta} - L \sin \theta \dot{\theta}^2 \end{aligned}$$

Inserting this in Eq. (19) we obtain :

$$\begin{aligned} L \cos \theta \ddot{\theta} - L \sin \theta \dot{\theta}^2 + \frac{(L \sin \theta)(L \cos \theta \dot{\theta})^2}{L^2 - L^2 \sin^2 \theta} + \frac{gL \sin \theta}{L^2} \sqrt{L^2 - L^2 \sin^2 \theta} &= 0 \\ L \cos \theta \ddot{\theta} - \cancel{L \sin \theta \dot{\theta}^2} + \frac{\cancel{L^3 \sin \theta \cos^2 \theta \dot{\theta}^2}}{L^2 \cos^2 \theta} + \frac{gL \sin \theta}{L^2} \times L \cos \theta &= 0 \\ L \cos \theta \ddot{\theta} + g \sin \theta \cos \theta &= 0 \end{aligned}$$

Dividing by $L \cos \theta$ both sides gives us back Eq. (6). So basically we have not done anything new so far. The new thing is that we have now the undermined multiplier by which we can calculate the constraint force.

Following Eq. (15) we identify $2\lambda x$ and $2\lambda y$ as constraint forces that acts along $\hat{x}$ and $\hat{y}$ respectively. So we can write force of constraint $\vec{F}_c$ as :

$$\vec{F}_c = 2\lambda x \hat{x} + 2\lambda y \hat{y} = 2\lambda \vec{r} \quad (20)$$

Combining Eqs. (18) and (19) we get :

$$\begin{aligned} 2\lambda &= -\frac{m\ddot{x} + \frac{m\dot{x}^2L^2}{L^2 - x^2}}{L^2 - x^2} - \frac{mg}{\sqrt{L^2 - x^2}} \\ &= -\frac{-mx\left(\frac{x\dot{x}^2}{L^2 - x^2} + \frac{gx}{L^2}\sqrt{L^2 - x^2}\right) + \frac{m\dot{x}^2L^2}{L^2 - x^2}}{L^2 - x^2} - \frac{mg}{\sqrt{L^2 - x^2}} \\ &= -\frac{-\frac{m\dot{x}^2L^2}{L^2 - x^2} - \frac{mgx^2}{L^2}\sqrt{L^2 - x^2} + \frac{m\dot{x}^2L^2}{L^2 - x^2}}{L^2 - x^2} - \frac{mg}{\sqrt{L^2 - x^2}} \end{aligned}$$

$$\begin{aligned}
&= -\frac{mx^2}{L^2 - x^2} + \frac{mgx^2}{L^2 \sqrt{L^2 - x^2}} - \frac{mg}{\sqrt{L^2 - x^2}} \\
&= -\frac{mx^2}{L^2 - x^2} - \frac{mg\sqrt{L^2 - x^2}}{L^2}
\end{aligned}$$

In terms of variable θ , we get :

$$2\lambda = -m \left[\dot{\theta}^2 + \frac{g}{L} \cos \theta \right] \quad (21)$$

Finally we have the force of constraint :

$$\vec{F}_c = 2\lambda \vec{r} = -m \left[\dot{\theta}^2 + \frac{g}{L} \cos \theta \right] \vec{r} \quad (22)$$

It's very clear from the sign that it acts radially inward, magnitude of which is simply :

$$F_c = \sqrt{(2\lambda x)^2 + (2\lambda y)^2} = 2|\lambda| \sqrt{x^2 + y^2} = 2|\lambda| L = m \left[L\dot{\theta}^2 + g \cos \theta \right] \quad (23)$$

This is what we have obtained naively by the application of Lagrange's equation of mixed type. But what is the physical interpretation of this constraint force which we will discuss next.

Physical Interpretation of Constraint force

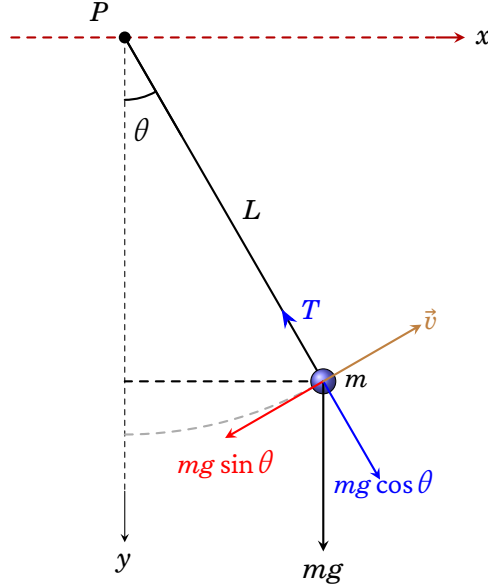


Figure 2: Simple Pendulum

As shown in Fig. 2, weight mg acts vertically downward. We can resolve this into two components, one is $mg \sin \theta$, that we all know as restoring force. Another component $mg \cos \theta$ that acts along the radial direction (i.e. along the string). As the string is inextensible the pendulum is forced to move on a circle of radius $r = L$. So there must be some forces of constraint present in the system, right! Our question is what's this force. It's very easy to guess that this constraint force is nothing but the Tension T of the string.

You might be tempted to write

$$T = mg \cos \theta,$$

but let me tell you this is a complete disaster. Now here is an interesting fact that we should keep in mind while dealing with this problem. As it is forced to move on a circular path, there is a centripetal acceleration that is acting radially along the string towards the suspension point. Tension must provide this. So from Newton's Law :

$$T - mg \cos \theta = ma_c \quad (24)$$

a_c is the centripetal acceleration form of which is given by :

$$a_c = \frac{v^2}{L} = L\dot{\theta}^2 \quad (25)$$

Using Eqs. (24) and (25) we get :

$$\boxed{T = mg \cos \theta + mL\dot{\theta}^2} \quad (26)$$

Which is exactly the same force of constraint we obtained in the previous section given in Eq. (23). This gives an physical explanation of this constraint force we have obtained through undermined multipliers just solving the EOM.

Motion is periodic, is it ?

Looking at the solution given by Eq. (8), it's very clear to see that the motion is periodic with the time period T :

$$T = \frac{2\pi}{\omega}$$

As, after time T the motion repeats itself, i.e.

$$\begin{aligned} \theta(t + T) &= A \sin \left[\omega \left(t + \frac{2\pi}{\omega} \right) \right] + B \cos \left[\omega \left(t + \frac{2\pi}{\omega} \right) \right] \\ &= A \sin [\omega t + 2\pi] + B \cos [\omega t + 2\pi] \\ &= A \sin [\omega t] + B \cos [\omega t] = \theta(t) \end{aligned}$$

Now that was very easy to say. But what about the generic equation given by Eq. (6). We multiply both sides of Eq. (6) by $\dot{\theta}$ to obtain :

$$\begin{aligned} \dot{\theta}\ddot{\theta} + \omega^2 \sin \theta \dot{\theta} &= 0 \\ \Rightarrow \frac{d}{dt} \left[\frac{\dot{\theta}^2}{2} - \omega^2 \cos \theta \right] &= 0 \end{aligned}$$

Integrating which one can obtain :

$$\boxed{\frac{\dot{\theta}^2}{2} - \omega^2 \cos \theta = \text{const}} \quad (27)$$

If you think a little bit, the first term is the kinetic term and second term is the potential one except a bit of prefactors which you can clearly see in the Lagrangian given in Eq. (4). So this is nothing but the total energy that appears on the L.H.S. and Eq. (27) says the total energy is conserved. This is expected as we don't see any explicit time dependence on the Lagrangian.

Now, consider the pendulum to be released from rest at an initial angle θ_0 . Since the pendulum is initially at rest, its kinetic energy is zero. As it begins to move, its potential energy decreases and is converted into kinetic energy, while the total mechanical energy remains conserved.

From Eq. (27), it is clear that $\dot{\theta}^2$ reaches its maximum when $\omega^2 \cos \theta$ is maximum, which occurs at $\theta = 0$. As the pendulum continues to move past this point, $\omega^2 \cos \theta$ decreases, and consequently so does $\dot{\theta}^2$. Eventually, at some angle ψ , the velocity becomes zero again. Therefore,

$$-\omega^2 \cos \psi = -\omega^2 \cos \theta_0 \implies \psi = -\theta_0^3$$

Thus, the pendulum moves from $\theta_0 \rightarrow 0 \rightarrow -\theta_0$. By the same argument, it subsequently retraces the corresponding path in the opposite direction $-\theta_0 \rightarrow 0 \rightarrow \theta_0$. Hence, the complete motion is

$$\theta_0 \rightarrow 0 \rightarrow -\theta_0 \rightarrow 0 \rightarrow \theta_0$$

after which the motion repeats itself. The motion is therefore **periodic**. This is illustrated in Fig. 3.

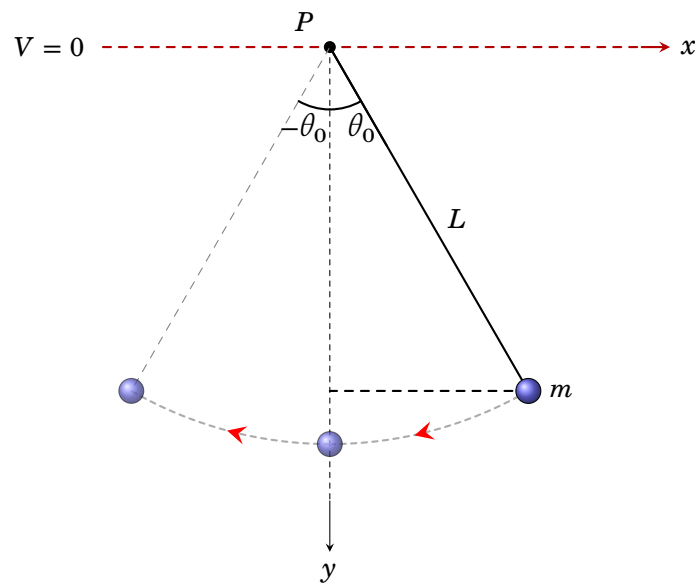


Figure 3: Periodic Motion of Simple Pendulum.

³We take the angle towards the right to be positive. Therefore, with this convention, an angle towards the left is negative.

The time period of the pendulum is given in terms of a complete elliptic integral of the first kind as

$$T = 4\sqrt{\frac{l}{g}} K\left(\sin \frac{\theta_0}{2}\right), \quad (28)$$

where the complete elliptic integral of the first kind is defined as

$$K(k) = \int_0^{\frac{\pi}{2}} \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}}. \quad (29)$$

Therefore, writing the elliptic integral explicitly, the period can be expressed as

$$T = 4\sqrt{\frac{l}{g}} \int_0^{\frac{\pi}{2}} \frac{d\phi}{\sqrt{1 - \sin^2\left(\frac{\theta_0}{2}\right) \sin^2 \phi}}. \quad (30a)$$

$$= 2\pi\sqrt{\frac{l}{g}} \sum_{n=0}^{\infty} \left[\frac{(2n)!}{2^{2n} (n!)^2} \right]^2 \sin^{2n} \left(\frac{\theta_0}{2} \right) \quad (30b)$$

$$= 2\pi\sqrt{\frac{l}{g}} \left[1 + \frac{1}{4} \sin^2 \frac{\theta_0}{2} + \frac{9}{64} \sin^4 \frac{\theta_0}{2} + \frac{25}{256} \sin^6 \frac{\theta_0}{2} + \frac{1225}{16384} \sin^8 \frac{\theta_0}{2} + \dots \right]^4 \quad (30c)$$

Here, θ_0 is the maximum angular displacement (amplitude) of the pendulum. Notice that the period depends on the amplitude θ_0 , unlike the familiar small-angle result. Following [Eq. \(30c\)](#), one can recover the small-angle result by neglecting higher order terms of $\mathcal{O}\left(\sin^2 \frac{\theta_0}{2}\right)$. i.e. taking approximation :

$$\theta_0 \ll 1 \implies \sin^2 \frac{\theta_0}{2} \approx 0.$$

⁴See, for example, G. B. Arfken and H. J. Weber, *Mathematical Methods for Physicists*, 5th ed., Academic Press (1995), Sec. 5.8, “Elliptic Integrals.”

Acknowledgments

I have studied this problem quite a few times. First during my bachelors then in masters. Again when I taught Classical Mechanics PH31207 in 2025 Autumn semester. During the tutorial session of Classical Mechanics PH31207 (Autumn 2026) that was held on 13/08/26, I realized the fact that we could do a better interpretation of this problem in a very nice way using the Lagrange's EOM. To my best knowledge in most of the textbooks I read on classical mechanics I had not found this connection. So I decided to write it up in a compact way. And that's the motivation behind this.

I would like to thank Dr. Aniket Patra who is the instructor of this course (Classical Mechanics PH31207 Autumn 2026) and also my friends Sandipan Hazra and Abhinaba Pahari for the helpful discussions.

And specially to those who never stop asking questions.

“There is no such thing as a stupid question.
Just stupid people who ask questions.”

Chris Berman

Type Setting Notes

- ★ This document has been typeset using $\text{\LaTeX}$.
- ★ The document was compiled using LuaLaTeX. The main text font is *Crimson Pro*, and the mathematical font is *TeX Gyre Schola Math*.
- ★ To use these fonts, include the following in the preamble:

```
\usepackage{fontspec}
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```
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```

```
\setmainfont{Crimson Pro}
```

```
% Mathematical font of the document
```

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\setmathfont{TeX Gyre Schola Math}
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